

SOUMIK SARKER ANIK

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SUMMARY

Master's graduate in Software Engineering with **three** years of research and industry experience in the field of machine learning and data science. Highly proficient in deep learning models and fuzzy logic systems. Co-author of three journal articles and two conference papers with a focus on predictive modeling and the analysis of complex geospatial, environmental, healthcare, and time series datasets. Fast learner, self-motivated, and tech enthusiast with excellent analytical and coding skills proven through work experience.

EDUCATION

Memorial University of Newfoundland 2023 – 2025
Masters of Applied Science, Software Engineering St John's, NL

North South University 2017 – 2021
Bachelor of Science, Computer Science – GPA 3.77/4.00 (Magna Cum Laude) Dhaka, Bangladesh

WORK EXPERIENCE

Online Data Analyst (Freelancer) May 2025 – Present

TELUS Digital

- Evaluating AI-generated content for accuracy, factual correctness, groundedness, and adherence to composition standards using internal guidelines.
- Categorizing and annotating search queries based on user intent to support large-scale supervised ML model training.
- Providing qualitative feedback to improve AI response logic and enhance model outputs for real-world user experience.

Machine Learning Engineer (Intern) Sep 2024 – Dec 2024

ExxonMobil Corporation

- Developed and implemented unsupervised ML models for anomaly detection in industrial machinery time series data.
- Performed necessary feature engineering and preprocessing steps on the data to enhance model performance.
- Utilized Azure Databricks and Azure Data Explorer (ADX) for data extraction and contributed to designing the solution architecture.

University Research Assistant Sep 2019 – Feb 2022

North South University

- Published three journal articles and two book chapters showcasing expertise in advanced ML algorithms across domains, with over 70 citations.
- Implemented different classification, clustering, and hybrid neural network models with my research team of five people.
- Conducted research in diverse domains, including healthcare analytics (COVID-19 transmission modeling), geospatial analysis (flood risk assessment), and transportation safety (accident severity prediction).

PROJECTS AND PUBLICATIONS

- **Flood Risk Assessment in Bangladesh** ([Publication](#) – [Geocarto International, 2022](#))
GIS-based national-scale flood risk modeling using hybrid DNN and fuzzy AHP methods.
- **Hybridized SVR for Flood Susceptibility Mapping** ([Publication](#) – [IEA/AIE 2021](#))
Applied Genetic Algorithm-optimized SVR models for flood susceptibility prediction using GIS data.
- **Road Crash Severity Prediction Using Machine Learning** ([Publication](#) – [ICCCI 2020](#))
Machine learning-based approach using AHC, Random Forest, and C5.0 models for crash prediction.
- **Time Series Anomaly Detection in Gas Compression Data** ([Project](#))
Developed unsupervised models (DBSCAN, Isolation Forest) to identify equipment anomalies in sensor data.

SKILLS

- **Programming Languages:** Python, R, SQL, C/C++, MATLAB, GIS Softwares (ArcGIS)
- **Machine Learning and NLP:** scikit-learn, Keras, TensorFlow, PyTorch, NLTK, NumPy, pandas
- **Data Visualization:** Matplotlib, Seaborn, Power BI
- **MLOps and DevOps:** CI/CD pipelines, MLflow, Git, Azure Databricks, Azure Data Explorer (ADX)
- **Domain Specific:** Healthcare Data Analytics, Geospatial Data Analysis, Hybrid ML models, Time Series Data Analysis, Predictive Modelling, Anomaly Detection